

Topic 25

Astronomy and Cosmology

Revision Booklet

This booklet covers:

- Luminosity and Radiant Flux Intensity
- Standard Candles and Distance Measurement
- Wien's Displacement Law and Stellar Temperature
- The Stefan–Boltzmann Law and Stellar Radii
- Redshift and the Expanding Universe
- Hubble's Law and the Big Bang

Luminosity and Radiant Flux Intensity

Luminosity

The **luminosity** L of a star is the total power of electromagnetic radiation emitted by the star in all directions.

$$L \quad \text{units: W (watts)}$$

Luminosity depends on the star's **surface temperature** and **surface area** — not on its distance from us.

Radiant Flux Intensity

The **radiant flux intensity** F (sometimes called apparent brightness) is the power of radiation received per unit area at a detector (e.g. a telescope on Earth).

$$F \quad \text{units: W m}^{-2}$$

F depends on both the luminosity of the source and its distance d .

Inverse Square Law for Radiant Flux

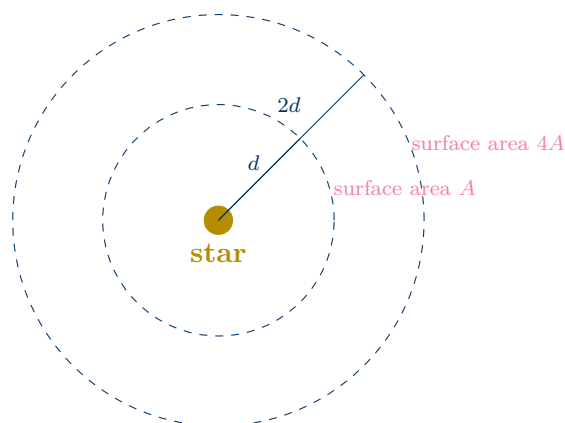
Assuming the star radiates uniformly in all directions and there is no absorption by the intervening medium, the radiation spreads over a sphere of area $4\pi d^2$:

$$F = \frac{L}{4\pi d^2}$$

F = radiant flux intensity at distance d (W m^{-2})

L = luminosity of the star (W)

d = distance from the star (m)



$F \propto 1/d^2$: double the distance, quarter the flux

Standard Candles

Standard Candle

A **standard candle** is an astronomical object whose **luminosity is known** (or can be determined independently of distance). By measuring the radiant flux intensity F received from the object and using $F = L/(4\pi d^2)$, its distance d can be calculated:

$$d = \sqrt{\frac{L}{4\pi F}}$$

Standard Candles Used in Practice

- **Cepheid variable stars:** pulsating stars whose *period of variation* is related to their luminosity (period–luminosity relation). Measure the period $\Rightarrow$ know $L \Rightarrow$ measure $F \Rightarrow$ find d .
- **Type Ia supernovae:** thermonuclear explosions of white dwarf stars that all reach approximately the same peak luminosity. Visible across enormous distances (billions of light-years), making them useful for measuring distances to distant galaxies.

Assumptions and Limitations

The inverse square law assumes:

- No absorption of radiation between source and detector (no dust or gas in the way).
- The source radiates **isotropically** (equally in all directions).

Dust absorption causes stars to appear **dimmer** than expected, which would lead to an **overestimate** of distance if uncorrected.

Wien's Displacement Law and Stellar Temperature

Black-Body Radiation

Stars approximate **black bodies** — objects that absorb all incident radiation and emit a characteristic continuous spectrum that depends only on temperature. The peak wavelength of this spectrum shifts with temperature.

Wien's Displacement Law

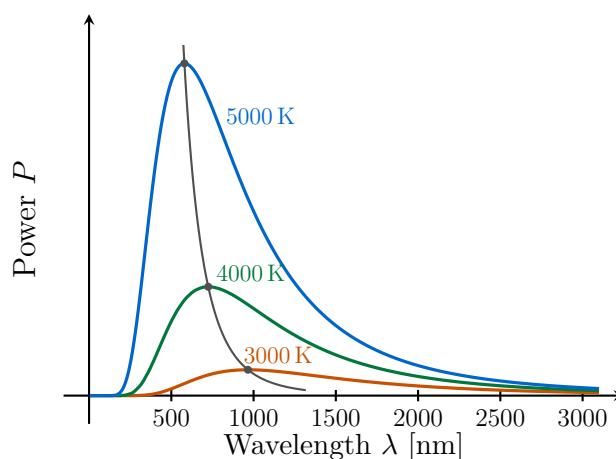
$$\lambda_{\max} \propto \frac{1}{T} \quad \text{equivalently} \quad \lambda_{\max} T = b$$

$\lambda_{\max}$ = wavelength at peak intensity of the spectrum (m)

T = surface temperature of the star (K)

b = Wien's constant = 2.90×10^{-3} m K

A **hotter** star has its peak at a **shorter** wavelength (bluer colour). A **cooler** star peaks at a **longer** wavelength (redder colour).



The Stefan–Boltzmann Law and Stellar Radii

Stefan–Boltzmann Law

For a black-body sphere (approximating a star) of radius r and surface temperature T :

$$L = 4\pi r^2 \sigma T^4$$

L = luminosity (W)

r = radius of the star (m)

σ = Stefan–Boltzmann constant = $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

T = surface temperature (K)

Estimating Stellar Radius

By combining Wien's law and the Stefan–Boltzmann law:

1. Measure λ_{max} from the star's spectrum $\Rightarrow$ use Wien's law to find T .
2. Measure F (radiant flux intensity) and find d (e.g. via a standard candle or parallax).
3. Find L from $L = 4\pi d^2 F$.
4. Rearrange Stefan–Boltzmann: $r = \sqrt{\frac{L}{4\pi\sigma T^4}}$

Using Ratios

Exam questions often ask you to *compare* two stars rather than calculate absolute values. In that case, form a ratio to cancel constants:

$$\frac{L_1}{L_2} = \frac{r_1^2 T_1^4}{r_2^2 T_2^4}$$

This avoids large numbers and reduces the risk of errors.

Redshift and the Expanding Universe

Redshift

When a source of light moves **away** from an observer, the observed wavelength is **longer** (shifted towards the red end of the spectrum) than the wavelength emitted. This is the **Doppler effect** applied to light.

The lines in the emission or absorption spectrum of a distant galaxy are observed at **longer wavelengths** than those of the same element measured in the laboratory.

Redshift Formula

For a source moving at speed $v \ll c$ relative to an observer:

$$z = \frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$$

z = redshift (dimensionless)

$\Delta\lambda$ = shift in wavelength (= observed λ – emitted λ) (m)

λ = emitted (rest-frame) wavelength (m)

Δf = shift in frequency (Hz)

v = recession speed of the source (m s^{-1})

Redshift as Evidence for Expansion

- Observations of distant galaxies show that their spectral lines are **all redshifted**.
- The greater the distance of a galaxy, the greater its redshift.
- This indicates that distant galaxies are moving **away** from us, and the further away they are, the faster they recede.
- This is consistent with the **Universe expanding**: it is space itself that is stretching, carrying galaxies apart, rather than the galaxies moving through space.

Hubble's Law and the Big Bang Theory

Hubble's Law

$$v \approx H_0 d$$

v = recession speed of a galaxy (m s^{-1})

H_0 = Hubble constant (s^{-1} , though often quoted in $\text{km s}^{-1} \text{Mpc}^{-1}$)

d = distance of the galaxy from Earth (m)

In CIE examinations, SI units are used: H_0 in s^{-1} and d in metres.

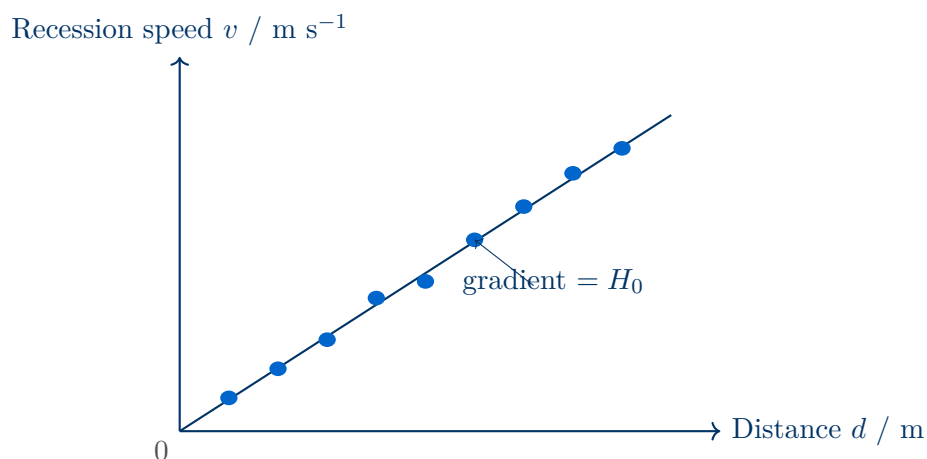
Hubble's Law and the Big Bang Theory

- Hubble's law states that the recession speed of a galaxy is proportional to its distance.
- If all galaxies are currently moving apart, then in the past they must have been **closer together**.
- Extrapolating back in time, all matter was once concentrated in an extremely hot, dense state — the **Big Bang**.
- An estimate of the **age of the Universe** can be obtained from:

$$t \approx \frac{1}{H_0}$$

(This assumes a constant rate of expansion, which is a simplification.)

Hubble Plot: Recession Speed vs Distance



Determining H_0 from a Graph

A graph of recession speed v against distance d gives a straight line through the origin. The **gradient** of this line is the Hubble constant H_0 . In practice, there is significant scatter due to the difficulty of measuring distances to distant galaxies accurately.

Formula Summary Sheet

Formula	Quantity	Units
$F = L/(4\pi d^2)$	Inverse square law / flux	W m^{-2}
$\lambda_{\text{max}}T = b$	Wien's displacement law	m K
$L = 4\pi r^2\sigma T^4$	Stefan–Boltzmann law	W
$\Delta\lambda/\lambda \approx \Delta f/f \approx v/c$	Redshift formula	—
$v \approx H_0 d$	Hubble's law	$\text{m s}^{-1}, \text{m}$
$t \approx 1/H_0$	Age of Universe estimate	s

Constants: $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}$, $b = 2.90 \times 10^{-3} \text{ m K}$, $c = 3.00 \times 10^8 \text{ m s}^{-1}$

Exam Technique and Problem-Solving Strategy

Step-by-Step for Stellar Radius Problems

1. Find T : measure λ_{max} from spectrum $\Rightarrow T = b/\lambda_{\text{max}}$.
2. Find L : use standard candle or parallax to get d ; then $L = 4\pi d^2 F$.
3. Find r : rearrange Stefan–Boltzmann: $r = \sqrt{L/(4\pi\sigma T^4)}$.

Common Errors — Avoid These!

- Forgetting the 4π in both $F = L/4\pi d^2$ and $L = 4\pi r^2\sigma T^4$.
- Confusing **luminosity** (intrinsic power, independent of distance) with **flux intensity** (observed brightness, depends on distance).
- Using $\Delta\lambda = \lambda_{\text{obs}} - \lambda_{\text{emitted}}$ but getting the **sign wrong**: for a receding source, $\Delta\lambda > 0$ (observed wavelength is longer).
- Applying $v = H_0 d$ with d in Mpc and H_0 in $\text{km s}^{-1} \text{Mpc}^{-1}$ — in CIE exams **always convert to SI** (metres and s^{-1}).
- Stating that the Big Bang means galaxies are moving **through space** — more accurately, **space itself is expanding**.

Worked Examples

Example 1 — Stellar Radius from Wien and Stefan–Boltzmann

Question: A star has a peak emission wavelength of 480 nm and a luminosity of 5.2×10^{26} W. Estimate its radius. ($b = 2.90 \times 10^{-3}$ m K, $\sigma = 5.67 \times 10^{-8}$ W m⁻²K⁻⁴)

Solution

Solution:

Step 1 — Surface temperature:

$$T = \frac{b}{\lambda_{\max}} = \frac{2.90 \times 10^{-3}}{480 \times 10^{-9}} = 6040 \text{ K}$$

Step 2 — Rearrange Stefan–Boltzmann:

$$r = \sqrt{\frac{L}{4\pi\sigma T^4}} = \sqrt{\frac{5.2 \times 10^{26}}{4\pi \times 5.67 \times 10^{-8} \times (6040)^4}}$$

$$T^4 = (6.04 \times 10^3)^4 = 1.33 \times 10^{15} \text{ K}^4$$

$$r = \sqrt{\frac{5.2 \times 10^{26}}{4\pi \times 5.67 \times 10^{-8} \times 1.33 \times 10^{15}}} = \sqrt{\frac{5.2 \times 10^{26}}{9.47 \times 10^8}} = \sqrt{5.49 \times 10^{17}} = 7.4 \times 10^8 \text{ m}$$

Example 2 — Distance Using Flux and Luminosity

Question: A type Ia supernova has a peak luminosity of 2.0×10^{36} W. It is observed with a radiant flux intensity of 3.5×10^{-14} W m⁻². Calculate its distance.

Solution

Solution:

$$d = \sqrt{\frac{L}{4\pi F}} = \sqrt{\frac{2.0 \times 10^{36}}{4\pi \times 3.5 \times 10^{-14}}}$$

$$d = \sqrt{\frac{2.0 \times 10^{36}}{4.40 \times 10^{-13}}} = \sqrt{4.55 \times 10^{48}} = 2.1 \times 10^{24} \text{ m}$$

Example 3 — Recession Speed and Age of Universe

Question: A galaxy shows a spectral line at 656 nm that is observed at 689 nm. The Hubble constant is $H_0 = 2.2 \times 10^{-18}$ s⁻¹. Calculate (a) the recession speed of the galaxy, (b) its distance, and (c) an estimate of the age of the Universe.

Solution

Solution:

$$(a) \frac{\Delta\lambda}{\lambda} = \frac{689 - 656}{656} = \frac{33}{656} = 0.0503$$

$$v = 0.0503 \times c = 0.0503 \times 3.00 \times 10^8 = 1.51 \times 10^7 \text{ m s}^{-1}$$

$$(b) d = v/H_0 = 1.51 \times 10^7 / 2.2 \times 10^{-18} = \mathbf{6.9 \times 10^{24} \text{ m}}$$
$$(c) t \approx 1/H_0 = 1/(2.2 \times 10^{-18}) = \mathbf{4.5 \times 10^{17} \text{ s}} (\approx 14 \text{ billion years})$$

Practice Exam Questions

Section A — Short Answer Questions

Q1. Define (a) luminosity and (b) radiant flux intensity. State the relationship between them and the distance d to the source.

[4 marks]

Q2. The Sun has a surface temperature of 5800 K. Calculate the peak wavelength of its emission spectrum and state what colour this corresponds to.

[2 marks]

Q3. Explain what is meant by a *standard candle* and describe one example of a standard candle used in astronomy.

[3 marks]

Q4. A hydrogen spectral line has a rest wavelength of 434 nm. In the spectrum of a distant galaxy it is observed at 461 nm. Calculate the recession speed of the galaxy.

[3 marks]

Q5. Explain how observations of redshift from distant galaxies lead to the conclusion that the Universe is expanding.

[3 marks]

Section B — Longer Structured Questions

Q6. Star A has surface temperature 12 000 K and radius 3.5×10^9 m. Star B has the same luminosity as Star A but a surface temperature of 4500 K.

(a) Calculate the luminosity of Star A.

[2 marks]

(b) Calculate the radius of Star B.

[3 marks]

(c) State which star would appear bluer and explain why.

[2 marks]

Q7. A galaxy is observed to have a recession speed of 4.8×10^6 m s⁻¹. The Hubble constant is $H_0 = 2.2 \times 10^{-18}$ s⁻¹.

(a) Calculate the distance of the galaxy.

[2 marks]

(b) A spectral line in the galaxy's spectrum has an emitted wavelength of 589 nm. Calculate the observed wavelength.

[2 marks]

(c) Explain how Hubble's law provides evidence for the Big Bang theory.

[3 marks]

Mark Scheme and Answers

Q1. (a) Luminosity: the total power of electromagnetic radiation emitted by a star (in all directions) [2]. (b) Radiant flux intensity: the power of radiation received per unit area at the observer's location [1]; $F = L/(4\pi d^2)$ [1].

Q2. $\lambda_{\max} = b/T = 2.90 \times 10^{-3}/5800 = \mathbf{500 \text{ nm}}$ [1]; this corresponds to green light (near the centre of the visible spectrum) [1].

Q3. A standard candle is an astronomical object of known luminosity [1]; example: type Ia supernova (all reach the same peak luminosity) [1]; or Cepheid variable (period of brightness variation gives luminosity via period–luminosity relation) [1].

Q4. $\Delta\lambda = 461 - 434 = 27 \text{ nm}$ [1]; $v = c\Delta\lambda/\lambda = 3.00 \times 10^8 \times 27/434$ [1] = $\mathbf{1.87 \times 10^7 \text{ m s}^{-1}}$ [1].

Q5. Spectral lines from distant galaxies are observed at longer wavelengths (redshifted) than expected [1]; this indicates the galaxies are moving away from us [1]; the greater the

distance, the greater the redshift/recession speed — consistent with all space expanding uniformly [1].

Q6(a). $L = 4\pi r^2 \sigma T^4 = 4\pi \times (3.5 \times 10^9)^2 \times 5.67 \times 10^{-8} \times (12000)^4$ [1]; $= 4\pi \times 1.225 \times 10^{19} \times 5.67 \times 10^{-8} \times 2.07 \times 10^{16} = \mathbf{1.80 \times 10^{29}}$ W [1].

Q6(b). Same L ; $r_B^2 T_B^4 = r_A^2 T_A^4$ [1]; $r_B = r_A (T_A/T_B)^2 = 3.5 \times 10^9 \times (12000/4500)^2 = 3.5 \times 10^9 \times 7.11$ [1] $= \mathbf{2.49 \times 10^{10}}$ m [1].

Q6(c). Star A is bluer [1]; it has a higher surface temperature, so by Wien's law its peak wavelength is shorter (towards the blue end of the spectrum) [1].

Q7(a). $d = v/H_0 = 4.8 \times 10^6 / 2.2 \times 10^{-18} = \mathbf{2.18 \times 10^{24}}$ m [2].

Q7(b). $\Delta\lambda = \lambda v/c = 589 \times 10^{-9} \times 4.8 \times 10^6 / 3.00 \times 10^8 = 9.42 \times 10^{-9}$ m [1]; observed $\lambda = 589 + 9.4 = \mathbf{598}$ nm [1].

Q7(c). Hubble's law shows recession speed is proportional to distance — the further away a galaxy, the faster it recedes [1]; this means all galaxies are moving apart from one another, implying the Universe is expanding [1]; extrapolating back in time, all matter must have originated from a single point — the Big Bang [1].

Revision Checklist

Use this checklist to track your understanding. Tick each box when you are confident:

Learning Objective	Confidence (1–3)
☐ Define luminosity and radiant flux intensity	
☐ Use the inverse square law $F = L/(4\pi d^2)$	
☐ Define a standard candle and explain how it is used to find distance	
☐ Give examples of standard candles (Cepheid variables, Type Ia supernovae)	
☐ State and apply Wien's displacement law $\lambda_{\max}T = b$	
☐ State and apply the Stefan–Boltzmann law $L = 4\pi r^2\sigma T^4$	
☐ Combine Wien and Stefan–Boltzmann to estimate stellar radius	
☐ Explain what redshift is and how it is observed in galaxy spectra	
☐ Use $\Delta\lambda/\lambda \approx v/c$ to find recession speeds	
☐ Explain how redshift provides evidence for an expanding Universe	
☐ State and apply Hubble's law $v \approx H_0d$ (SI units)	
☐ Use $t \approx 1/H_0$ as an estimate for the age of the Universe	
☐ Explain how Hubble's law leads to the Big Bang theory	

Key: 1 = Need more work 2 = Getting there 3 = Confident

Good luck with your revision!

Cosmology asks the biggest questions in physics — and answers them with the same tools you've used all year. A few formulas, careful unit conversions, and the ability to interpret a graph are all you need to explore the scale of the Universe itself.